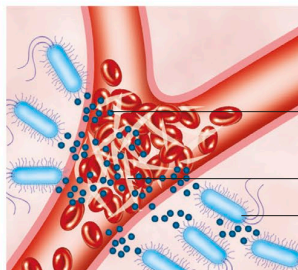


HOW BACTERIA CAUSE DAMAGE

Disease-causing bacteria can enter the body in several ways: via the airways or digestive tract, during sexual contact, or through broken skin. Some bacteria adhere to and invade body cells. Others produce poisonous substances called bacteriotoxins, or toxins, which may disrupt cell function. For example, the diphtheria toxin from the bacterium *Corynebacterium diphtheriae* damages heart muscle by inhibiting protein production. Some toxins are highly dangerous. A bucket of nerve toxin from *Clostridium botulinum* could kill everyone in the world.



Released toxins

Clot forming

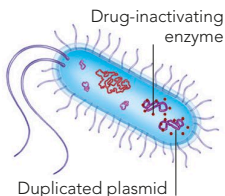
Bacterium

LEAKING VESSELS

Some bacteria release toxins that cause blood to clot in small blood vessels, depriving tissues and organs of their normal blood supply.

1 ROLE OF PLASMIDS

A plasmid may cause a bacterium to make enzymes against antibiotics, or to alter its surface receptors, where antibiotics bind. Then the plasmid duplicates itself.



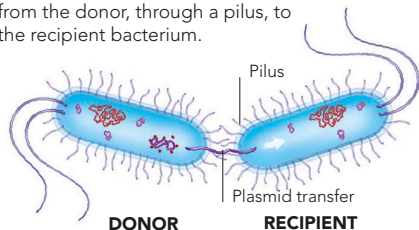
Duplicated plasmid

RESISTANCE TO ANTIBIOTICS

Many bacteria become resistant to antibiotics by changing (mutating) into new strains. Their most effective mechanism is the transfer of plasmids – fragments of the genetic material DNA – between bacteria. The gene for antibiotic resistance crops up by accident, and the bacterium possessing it can pass it to others.

2 PLASMID TRANSFER

Plasmid transfer takes place during a process known as conjugation. The plasmid copy is passed from the donor, through a pilus, to the recipient bacterium.

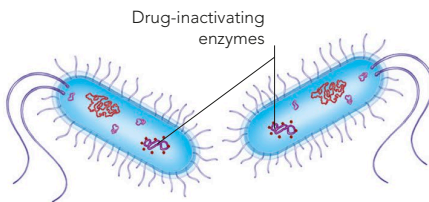


DONOR

RECIPIENT

3 DRUG-RESISTANT STRAINS

Recipient bacteria inherit the resistant gene. Plasmid transfer produces populations of bacteria resistant to a range of antibiotics.



SUPERBUGS

Some bacteria pass through their life cycle in less than 20 minutes. Fast reproduction, coupled with the incredible numbers of bacteria and rapid transfer of genetic information, gives great scope for mutation (see above). Many strains of bacteria resistant to wide-acting, or broad-spectrum, antibiotics have appeared. These so-called "superbugs" may not be resistant to more specialist, narrow-spectrum, antibiotics.

MRSA

Staphylococcus aureus bacteria resistant to the antibiotic meticillin are known as MRSA, and are a cause of concern in hospitals.

